

MSTAR PRACTICE WORKSHEET**Lesson 7-4: Compare and Order Integers and Rational Numbers**

Grade 6 Mathematics · Standard 6.NOS.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which statement is TRUE?

- ☒ A $-5 < -2$
- ☐ B $-4 > 0$
- ☐ C $-3 > -1$
- ☐ D $-1 < -6$

-5 is farther to the left on the number line than -2 , so $-5 < -2$ is true.

Question 2 SELECTED RESPONSE — CORRECT: AWhich integer is the LEAST: -3 , 5 , -8 , 2 , 0 ?

- ☒ A -8
- ☐ B -3
- ☐ C 0
- ☐ D 5

-8 is farthest to the left on the number line, making it the least (smallest) integer.

Question 3 SELECTED RESPONSE — CORRECT: B

What is $|-8|$?

- ☐ A 0
- ☒ B 8
- ☐ C -8
- ☐ D -16

Absolute value is distance from 0, never negative (zero is allowed). $|-8| = 8$.

Question 4 SELECTED RESPONSE — CORRECT: C

Three cities have temperatures of -12°F , -5°F , and 3°F . Which city is warmest?

- ☐ A -5°F
- ☐ B -12°F
- ☒ C 3°F
- ☐ D They are equal

3°F is the greatest because it is farthest to the right on the number line. The order from coldest to warmest is -12, -5, 3.

Question 5 SELECTED RESPONSE — CORRECT: A

Four spots: -3 m, 2 m, -7 m, 0 m. Order them from least to greatest.

- ☒ A -7, -3, 0, 2
- ☐ B -7, -3, 2, 0
- ☐ C -3, -7, 0, 2
- ☐ D 2, 0, -3, -7

The deepest spot is the cave at -7 (farthest left), then the tide pool -3, then the boat deck 0, then the flag at 2. Least to greatest: -7, -3, 0, 2.

Question 6

SELECTED RESPONSE — CORRECT: D

Three buried chests sit at -9 m, -4 m, and -15 m below sea level. Which chest is at the GREATEST elevation (closest to the surface)?

- ☐ A -9 m
- ☐ B -15 m
- ☐ C They are equal
- ☒ D -4 m

Closer to zero means greater for negatives. -4 is only 4 below the surface, so -4 is the greatest of -9 , -4 , and -15 .

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: C

Order these numbers from LEAST to GREATEST: -2 , 1.5 , -4.5 , 0 , 3 .

- ☐ A -2 , -4.5 , 0 , 1.5 , 3
- ☐ B 3 , 1.5 , 0 , -2 , -4.5
- ☒ C -4.5 , -2 , 0 , 1.5 , 3
- ☐ D -4.5 , -2 , 0 , 3 , 1.5

Reading a number line from left to right gives -4.5 , -2 , 0 , 1.5 , 3 .

- **A:** The two negatives are reversed. Negative 4.5 sits further left than negative 2, so it comes first.
- **B:** That is greatest to LEAST. The question asks for least to greatest.
- **D:** The last two are reversed. 1.5 is smaller than 3.

Part B — correct answer: B

Which statement correctly compares -4.5 and -2 ?

- ☐ A $-4.5 > -2$, because $4.5 > 2$.
- ☒ B $-4.5 < -2$, because -4.5 lies further left on the number line.
- ☐ C They are equal in value.
- ☐ D $-4.5 > -2$, because it is further from zero.

Left is less. -4.5 sits to the left of -2 , so -4.5 is the smaller number even though its absolute value is larger.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL comparisons that are TRUE:

☒ A $-7 < -2$

☒ B $0 > -1$

☒ C $-0.5 > -1.5$

☐ D $-10 > -3$

☒ E $2.5 > -9$

A, B, C, and E are true. D is false — -10 lies further left than -3 , so $-10 < -3$.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A student says: 'Negative numbers are always less than positive numbers, so -100 is less than 1 .' Is this correct? Now the student says: ' -100 is less than -1 because 100 is greater than 1 .' Is THIS correct? Explain the difference.

Model answer: The first statement is true: -100 is negative and 1 is positive, so -100 sits left of 1 on the number line and is less. The second answer, $-100 < -1$, is also true, but the reason given is wrong — it is not because $100 > 1$, it is because -100 lies 100 units below zero while -1 lies only 1 unit below, putting -100 farther left.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.